Total No. of Questions : 9] [Total No. of Printed Pages : 4

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F-404

B.C.A. (First Semester)

Examination, 2023-24

(New Course)

Paper - BCA-104-N

MATHEMATICS - 1

Time : Three Hours] *[Maximum Marks : 70*

Note : Attempt any five questions. **All** questions carry equal marks. Symbols used have their usual meanings.

1. (a) Compute $\lim_{x\to 0}\frac{e^{\sin x}-1}{x}$

(b) Show that $f(x) = \log x$, $x\in(0,\infty)$ is continuous at every point of its domain.

2. (a) Find the n^{th} derivative of $y=\tan^{-1}(x/a)$.

(b) If $y=\cos(m\sin^{-1}x)$, show that

$(1-x^2)y_{n+2}-(2n+1)xy_{n+1}+(m^2-n^2)y_n=0$

3. **(a)** If $x^y y^x=c$, show that at $x=y=z$

$\frac{\partial^2 z}{\partial x\partial y} = -(x\log e x)^{-1}$

(b) If $u=xy+yz+zx$, $v=x^2+y^2+z^2$, and $w=x-y+z$, show that they are functionally related and find the relation between them.

4. **(a)** Evaluate $\int_0^{\pi/6}\cos^3 3\theta\sin^3 6\theta d\theta$

(b) Find the perimeter of the circle $x^2+y^2=a^2$.

5. **(a)** Evaluate $\int_0^1\int_0^{x^2}(x^2+y^2)dx\,dy$

(b) Evaluate $\int_0^a\int_0^{a-x}\int_0^{a-x-y}x^2\,dx\,dy\,dz$

6. (a) Solve $(x+1)\frac{dy}{dx} - y = e^x(x+1)^2$

(b) Solve $\frac{d^2y}{dx^2} - 4\frac{dy}{dx} + 3y = e^{2x} + \cos x$

7. (a) A particle moves along the curve $x=t^3+1$, $y=t^2$, $z=2t+5$, where t is the time. Find the components of its velocity and acceleration at t=1 in the direction $2i+3j+6k$.

(b) If $\vec{a} = x^2yz\,\hat{i} - 2xz^3\hat{j} + xz^2\hat{k}$ and $\vec{b} = 2z\,\hat{i} + y\hat{j} - x^2\hat{k}$, find the value of $\frac{\partial^2}{\partial x^2}(\vec{a}\times\vec{b})$ at the point (1,0,1).

8. (a) Find the value of n for which the vector $r^n\vec{r}$ is solenoidal, where $\vec{r} = x\,\hat{i} + y\hat{j} + z\hat{k}$

(b) Prove that, for every field $\vec{v}$; div curl $\vec{v} = 0$.

9. (a) The cost of fencing a circular field at the rate of Rs. 24 per metre is Rs. 5280. The field is to be ploughed at the rate of Rs. 0.50 per m^2. Find the cost of ploughing the field ($\pi = 22/7$).

(b) Find the equation of a line making an angle of 150° with the x-axis and cutting off an intercept 2 from y-axis.